

Requirement ID	Requirement	Verification Method	Procedure	Acceptance Criteria
STK-008	The system shall be transportable by one person.	Inspection	TP-001	One person can lift, carry, and deploy the transport configuration without disassembly beyond normal setup actions.
SYS-INT-003	The system shall support Ethernet communications.	Test	TP-011	The GCS establishes Ethernet connectivity and successfully transfers data.
SYS-INT-004	The system shall support USB peripheral connections.	Test	TP-011	Configured USB peripherals enumerate and operate without errors.
SYS-ENV-004	The system shall tolerate transportation-induced vibration.	Test	TP-007	After the transportation-vibration profile, hardware remains secure and passes post-test functional checks.
SYS-SEC-001	The system shall authenticate operator access.	Test	TP-010	Protected operator functions are unavailable until valid authentication is completed.
SYS-SEC-002	The system shall protect mission data.	Inspection	TP-010	Configured operating-system or application controls restrict

				unauthorized mission-data modification.
SYS-SEC-004	The system shall maintain audit logs of mission activities.	Test	TP-010	Configured mission and access events appear in the audit log with usable timestamps.
SYS-MIS-005	The system shall display the planned flight route to the operator.	Demonstration	TP-011	The complete planned route is displayed on the GCS map before launch.
SYS-DAT-003	The system shall assign a unique identifier to each mission data set.	Test	TP-002	Each generated mission data set has a unique identifier not reused by another data set.
SYS-DAT-004	The system shall preserve recorded mission data following a normal system shutdown.	Test	TP-002	Recorded mission data remains intact and readable after normal shutdown and restart.
SYS-DAT-005	The system shall permit authorized users to retrieve recorded mission data after mission completion.	Demonstration	TP-002	An authenticated user retrieves the selected completed-mission data set.
SYS-GCS-002	The GCS shall display vehicle position on a digital map.	Test	TP-011	The vehicle position symbol is displayed and updates on the digital map.
SYS-GCS-003	The GCS shall display vehicle altitude, heading, speed, and flight mode.	Test	TP-011	Altitude, heading, speed, and flight mode are simultaneously displayed and

				update during operation.
SYS-GCS-004	The GCS shall display battery status and available system-health information.	Test	TP-011	Battery and configured system-health indicators are visible and update during operation.
SYS-PWR-006	The system shall prevent reverse-polarity connection at externally accessible power interfaces through connector design or protective circuitry.	Inspection	TP-012	Connector keying or protective circuitry prevents damaging current flow under reversed-polarity test conditions.
SYS-HMI-001	The system shall present critical warnings in a visually distinguishable format.	Demonstration	TP-011	Critical warnings are visually distinct from normal status and advisory information.
SYS-HMI-002	The system shall identify the current vehicle flight mode to the operator.	Test	TP-011	The displayed flight mode matches the active vehicle mode during each test transition.
SYS-HMI-003	The system shall display the status of the command-and-control link.	Test	TP-011	The command-and-control link status changes correctly for connected, degraded, and lost conditions.
SYS-HMI-004	The system shall display the status of the telemetry link.	Test	TP-011	The telemetry-link status changes correctly for connected, degraded, and lost conditions.

SYS-HMI-005	The system shall display the status of the live-video link.	Test	TP-011	The live-video link status changes correctly for connected, degraded, and lost conditions.
SYS-HMI-006	The system shall require operator confirmation before executing a mission-termination command.	Test	TP-011	A mission-termination command is not executed until the operator completes the confirmation action.
SYS-MNT-004	The system shall provide externally visible identification for major electrical interfaces.	Inspection	TP-001	All designated major external electrical interfaces have legible identification.
SYS-MNT-006	The system shall permit inspection of major structural and electrical connections during scheduled maintenance.	Inspection	TP-001	All designated major structural and electrical connections are accessible for visual inspection.
SYS-CYB-001	The system shall restrict administrative configuration functions to authorized users.	Test	TP-010	Unauthorized users cannot access administrative configuration functions.
SYS-CYB-002	The system shall require authentication before access to protected GCS functions.	Test	TP-010	Protected GCS functions remain inaccessible until valid authentication is completed.
SYS-CYB-003	The system shall protect stored mission data from unauthorized modification using	Test	TP-010	An unauthorized account cannot modify protected

	available operating-system or application controls.			stored mission data.
SYS-CYB-006	The system shall record security-relevant access or configuration events when supported by the selected software.	Test	TP-010	Configured access and configuration events are recorded with event type and timestamp.